SMARTBRIDGE

SKILL WALLET — A SmartBridge Product

Scalability & Future Plan

Date	29-10-2026	**Team ID**	en25cs3030039DS
Project Name	Global Fuel Price and Taxation Trends	**Maximum Marks**	1 Mark

This document outlines how the current Global Fuel Price and Taxation Trends analytical system can be scaled to handle larger user bases, expanded historical datasets, and real-time market feeds, alongside the future technology roadmap.

CURRENT SYSTEM LIMITATIONS

S.No	Limitation	Impact	Priority
1	Static CSV Data Ingestion	Manual updates required for post-2026 fuel price and taxation data releases.	High
2	Single Small Snowflake Warehouse	Simultaneous heavy analytical queries from 50+ analysts could cause minor queue delays.	Medium
3	Tableau Public Public Domain Hosting	Lack of row-level security (RLS) for proprietary energy corporate pricing data.	Medium

SCALABILITY PLAN

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	**User Load**	Supports up to 20 concurrent Tableau viewers.	Implement multi-cluster auto-scaling Snowflake warehouses (COMPUTE_WH scale-out to 4 clusters).
2	**Data Storage**	12,450 records (~4.2 MB compressed in Snowflake).	Transition to automated S3 / Snowflake Pipe live streaming for real-time daily global fuel feeds.
3	**Performance**	Sub-second query response via result cache reuse.	Implement Materialized Views for pre-aggregated regional tax and Brent crude index calculations.
4	**Security**	Standard role-based access control (ACCOUNTADMIN/PUBLIC).	Deploy Snowflake Row-Level Security (RLS) and dynamic data masking for corporate tax data.

FUTURE ROADMAP

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Automated daily ingestion pipelines from Global Energy APIs (EIA, OPEC, World Bank).	Q3 2026	Real-time daily fuel price tracking.
Phase 3	Predictive Machine Learning models for 12-month Brent Crude and petrol pump price forecasting.	Q4 2026	Proactive fuel hedging insights.
Phase 4	Enterprise Tableau Server migration with personalized user alerts for tax rate threshold breaches.	Q1 2027	Automated executive notification.